

Semiconductor Device Simulation

A complete guide: written for a first-time reader with no technical background, and usable as a full technical reference after that.

This document is in two parts. **Part I** explains the project in plain language, assuming nothing beyond a first read. **Part II** is the full technical reference — the same equations, code structure, and results, in depth, for anyone who wants the underlying detail behind Part I's explanations. Every number in this document comes from an actual simulation run recorded in the project's own `docs/validation.md`; nothing is invented or illustrative.

PART I — PLAIN-LANGUAGE EXPLANATION

What is a transistor, and why simulate one?

A **transistor** is a tiny electrical switch. Almost every electronic device — phones, laptops, calculators — is built from billions of them wired together. The type studied in this project is called a **MOSFET** (metal-oxide-semiconductor field-effect transistor): apply a voltage to one terminal (the *gate*), and it controls whether current can flow between the other two terminals (the *source* and *drain*) — like a valve controlled by voltage instead of a hand.

Before a transistor is physically manufactured (an expensive, slow process), engineers **simulate** it on a computer: build a virtual model of its physical structure and geometry, and use software that solves the underlying physics equations to predict how it will behave. That is exactly what this project does — it never touches real silicon. Everything here is computed.

What question is this project actually trying to answer?

A transistor's design involves trade-offs. Three of its physical dimensions were studied here:

- **Channel length** — how long the path is that current travels through the device.
- **Gate oxide thickness** — how thick the insulating layer is between the gate and the channel underneath it.
- **Channel doping** — how heavily the silicon under the gate is chemically treated ("doped") with impurity atoms, which changes how easily current flows.

The question: as each of these is changed, what happens to (a) how much current the device delivers when switched on (**drive current**), (b) how much current leaks through when it should be fully off (**leakage current**), and (c) how sharply it switches between on and off? This matters directly for **low-power design** — devices like phones and wearables need transistors that switch cleanly while wasting as little power as possible, especially when "off."

How was it simulated, in plain terms?

Two ingredients:

- **DEVSIM** — free, open-source simulation software built specifically for semiconductor devices. It is given a description of the device's physical shape and material properties, and it numerically solves the equations of physics (electric fields, how electrons and "holes" — the absence of an electron, which acts like a positive charge carrier — move through the material) to predict its electrical behavior.
- **This project's own Python code** — everything around DEVSIM: describing the device's shape, running it at different voltages, reading out the results, and turning raw numbers into the standard measurements engineers use to compare transistors.

Before trusting DEVSIM for the (more complex) transistor, it was first tested on a simpler, well-understood device — a **PN junction** (a basic two-terminal diode, the building block a transistor is made from). Its simulated behavior was checked against a known, hand-calculated formula. They matched almost exactly — confirming DEVSIM's physics engine is trustworthy before using it on the transistor.

What was actually built and run, step by step

- **Step 1 — environment setup.** A reproducible software environment (so the exact same simulation can be re-run identically on another machine) with DEVSIM and the necessary Python tools installed and proven to work.
- **Step 2 — validate the physics.** Simulate the PN junction diode, compare against the textbook formula. It matched to an error of about 0.00000000000007% — effectively exact.
- **Step 3 — build the transistor.** A 2D cross-section of a real planar MOSFET structure: gate, oxide, source, drain, and the silicon body, all with realistic material properties. Run it at different gate and drain voltages and confirm it behaves like a transistor should — current flows only once the gate voltage is high enough, and increases smoothly and predictably after that.
- **Step 4 — vary the three design parameters.** Rebuild the transistor three separate times, each time changing exactly one of channel length, oxide thickness, or doping, and re-measure its behavior each time — the actual experiment this project set out to run.
- **Step 5 — score the trade-off.** Combine the results from Step 4 into one ranked comparison, to see which combination of settings looks best for a low-power design.
- **Step 6 — check the results are trustworthy.** Re-run the baseline transistor with a finer computational grid ("mesh") to see how much the answer changes just from computational precision, not physics — a standard way of quantifying how much to trust a number coming out of a simulation.

What was found, in plain terms

A thinner insulating oxide layer and lighter doping both let the transistor carry more current when switched on, and switch on more sharply — a clear, consistent, and expected result. **A shorter channel also increases the on-current**, again as expected from basic device theory.

None of the three parameters produced a measurable change in leakage (off-state) current in this simulation. That is not a bug and not a disappointing result — it is itself a finding, and an honest one: this project deliberately used a simplified, idealized transistor model, and that model does not include the physical mechanisms (present in real, very small modern transistors) that would make leakage current depend on these parameters. So the project's own data can answer "which settings maximize drive current" with confidence, but cannot answer "which settings minimize leakage" — and that limitation is stated directly rather than glossed over.

A combined "low-power score" was still computed to rank the seven configurations tested, but because the leakage-current measurements were too close together to be meaningful (their differences were smaller than the simulation's own numerical precision), that combined score ends up mostly reflecting drive current and switching sharpness rather than a genuine drive-vs-leakage trade-off. This is explained in full, with the reasoning shown, in Part II.

How much can these numbers be trusted?

Every simulation has some level of numerical approximation, the same way a finite-element structural simulation or a weather forecast does. This project measured its own precision directly rather than assuming it: re-running the baseline device with a twice-as-fine computational grid changed the key "switch-on sharpness" measurements by under 1%, but changed the on-current figure by about 5.7%. That means: trust the on/off switching behavior to high precision, and trust the exact on-current numbers to roughly one part in twenty, not further. This kind of self-check is standard good practice in computational engineering, and its result is reported honestly here rather than hidden.

A short glossary

Term	Plain-language meaning
MOSFET	The type of transistor studied — a voltage-controlled switch.
Gate / Source / Drain	The three terminals of the MOSFET: the gate controls the switch; current flows between source and drain.
Doping	Deliberately adding small amounts of other elements to silicon to control how well it conducts and what type of charge carrier dominates.
N-type / P-type	Two flavors of doped silicon: N-type conducts mainly via electrons (negative), P-type mainly via "holes" (effectively positive).
Channel	The thin region under the gate where current actually flows when the device is switched on.
Gate oxide	The thin insulating layer between the gate and the channel.
Drive current (I_{ON})	How much current flows when the device is fully switched on — higher is generally better for speed.
Leakage current (I_{OFF})	How much current still sneaks through when the device is supposed to be fully off — lower is better for power efficiency.
Threshold voltage (V_{TH})	The gate voltage at which the device starts turning on.
Subthreshold swing (SS)	How sharply the device switches from off to on; lower (fewer millivolts per decade of current) is a sharper, more efficient switch.
Mesh	The grid of points the simulation software uses to numerically solve the physics equations across the device's shape; a finer mesh is more accurate.
Validation	Checking a simulation's output against an independent, trusted reference (here, a hand-derivable formula) to confirm it is produced correctly.

Questions likely to come up, answered simply

Below: the questions most likely to be asked about a project like this, each answered in plain language first, with a pointer to the exact technical section in Part II for anyone who wants the full derivation or number.

Question	Plain-language answer
What did you actually build?	A computer simulation of a MOSFET transistor's physical structure, plus code that drives that simulation, measures its
Why should anyone trust a simulation instead of a real chip?	Because this simulation engine (DEVSIM) is an established, open-source tool, and this project independently checked
What's the headline finding?	Thinner gate oxide, lighter channel doping, and shorter channel length all increase drive current and switching sharpn
Why didn't leakage current change?	Because the transistor model used here is intentionally simplified and does not include the physical effects (present in
How do you know the results are numerically accurate, not just plausible-looking?	By corroborating the simulation with a much finer computational grid and measuring how much the answer moved.
Is this a real, sellable chip design?	No. It is a generic, simplified, physically-parameterized model used to study a general design trade-off, not a model of
What would you need to add to answer the leakage-related question properly?	A more detailed transistor model that includes short-channel leakage effects (specific physical mechanisms not includ
What tool was used to actually run the physics simulations?	DEVSIM, an open-source device simulation engine that numerically solves the standard semiconductor equations (ele

PART II — FULL TECHNICAL REFERENCE

Everything below is the detailed technical content behind Part I: the physics equations actually solved, how the code is organized, and the full numerical results for every stage of the project.

1. Scope and Research Question

Research question: how do channel length, gate oxide thickness, and channel doping influence the trade-off between drive current, leakage current, and switching behavior in a low-power context?

Device model: a generic, physically-parameterized planar silicon MOSFET — not a model of any specific commercial fabrication process. Every parameter in every result is labeled as one of: simulation input, assumed, analytically derived, or calibrated.

Out of scope, deliberately: 3D devices (FinFET/GAA), quantum transport, Monte Carlo transport, fabrication-process simulation, dependency on a real commercial process, more than a 3-variable parameter sweep, machine-learning-driven optimization, thermal/multiphysics simulation, device-circuit co-simulation.

2. The Physics Being Solved

Poisson's equation (electrostatics): relates electric potential to charge density.

$$\nabla \cdot (\epsilon \nabla \psi) = -q(p - n + N_D^+ - N_A^-)$$

Continuity equations (carrier conservation), with generation $G \approx 0$ and recombination R modeled via Shockley-Read-Hall (SRH):

$$\partial n / \partial t = (1/q) \nabla \cdot J_n + G - R$$

Drift-diffusion transport: carriers move under the electric field (drift) and their own concentration gradient (diffusion), linked by the Einstein relation $D = \mu V_T$.

$$J_n = q \mu_n n E + q D_n \nabla n$$

MOSFET threshold voltage (reference formula, used only as a sanity check, never as the reported result):

$$V_{TH} = V_{FB} + 2\phi_F + \sqrt{(4\epsilon_{Si}\epsilon_0 q N_A \phi_F) / C_{ox}}$$

All constants (q , k_B , ϵ_0 , ϵ_{Si} , ϵ_{ox} , n_i) and full derivations are in docs/physics.md.

3. Code Architecture

Folder	What it does
src/physics/	Physical constants, carrier-concentration formulas, doping-profile math, and the closed-form PN-junction formulas used to validate DEVSIM
src/device/	Builds the MOSFET's 2D mesh, regions, and contacts directly with DEVSIM's own meshing primitives, and generates its 2D doping profile
src/simulation/	Drives DEVSIM: sets up physics models, solves potential-only then full drift-diffusion, runs one full characterization sweep per device config
src/extraction/	Pure-NumPy functions turning a raw I_D-V_G sweep into the standard MOSFET figures of merit: V_TH, g_m, SS, I_ON, I_OFF.
src/optimization/	Combines I_ON, I_OFF, and g_m from every swept configuration into one weighted, normalized low-power trade-off score.
simulations/	One driver script per experiment — thin scripts that call into src/ and write results.
results/	Generated output: raw/ (per-run CSVs), processed/ (summary metrics tables), figures/ (plots), tables/. Regenerated by simulations/, not ha
tests/	Unit tests for the physics formulas, doping/extraction/optimization code.
docs/	project_manual.md, physics.md, roadmap.md, validation.md, limitations.md, optimization.md, HANDOFF.md.

Mental model: src/device/ builds the device, src/simulation/ solves it and sweeps bias, src/extraction/ turns the raw sweep into meaningful figures of merit, and src/optimization/ compares several devices. src/physics/ is what every result is checked against.

4. Results, Stage by Stage

4.1 PN junction — validating the solver

A 1D step junction ($N_A = N_D = 1e16 \text{ cm}^{-3}$) was solved in DEVSIM and compared against the closed-form built-in-potential formula: relative error 7.2×10^{-14} (tolerance 0.1%). Forward-bias ideality factor ≈ 1.01 , matching ideal diode theory.

Non-obvious finding: the first attempt failed the gate with a 0.136% error, purely because the analytical formula used CODATA-precise constants while DEVSIM uses its own slightly rounded internal constants. Switching to DEVSIM's exact constants fixed it — an apparent physics mismatch turned out to be a units/constants mismatch.

4.2 Baseline MOSFET

Planar NMOS ($L = 1 \mu\text{m}$, $t_{\text{ox}} = 10 \text{ nm}$, $N_A = 1e17$, $N_D = 1e20 \text{ cm}^{-3}$). Five sanity checks all passed: monotonic I_D - V_D and I_D - V_G , no divergence, I_{ON} ($5.31e-3 \text{ A/cm}$) above I_{OFF} ($3.26e-11 \text{ A/cm}$), and mid-channel surface electron density rising from $9.2e6$ to $9.6e17 \text{ cm}^{-3}$ under gate bias — confirming an actual inversion channel, not just a monotonic terminal current.

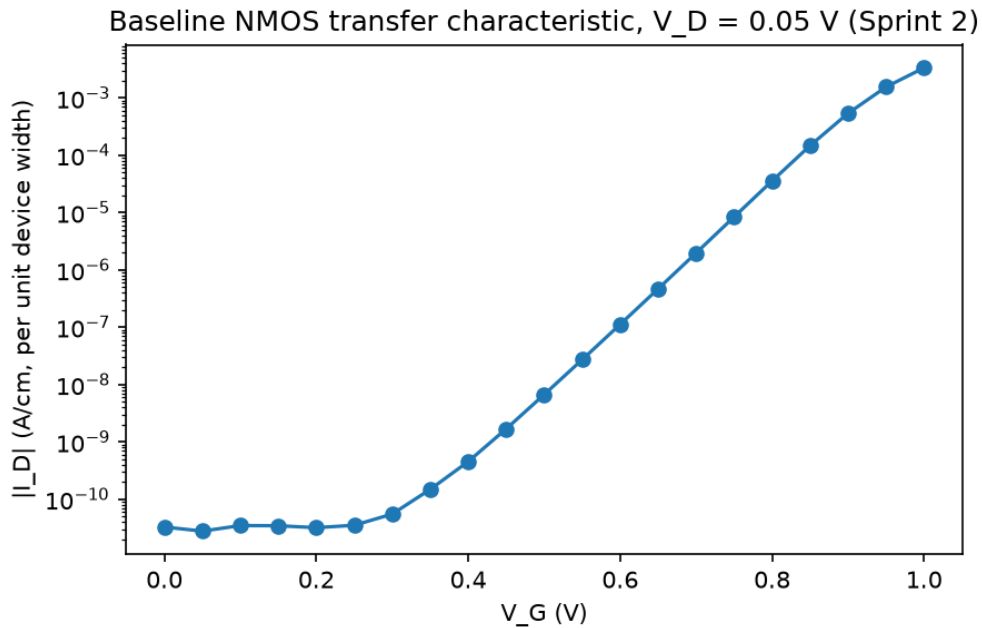


Figure 1. Baseline I_D - V_G (log scale): subthreshold exponential into strong inversion across roughly 11 decades.

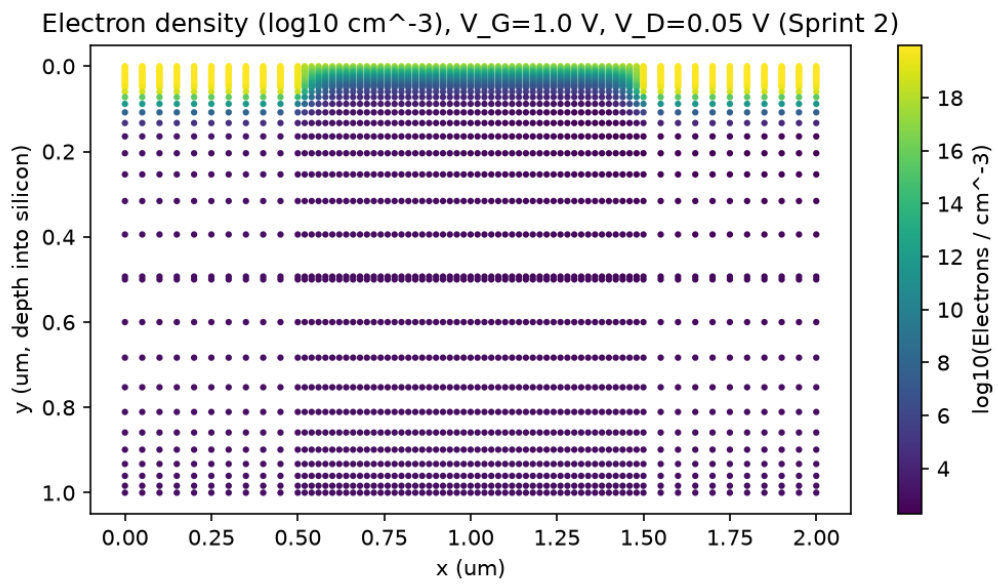


Figure 2. Electron density at $V_G = 1.0 \text{ V}$: inversion channel connects the source and drain at the surface.

4.3 Parameter sweeps

Sweep	Direction	Finding
Channel length L	5.0e-5 → 2.0e-4 cm	I_ON strictly decreases ($1.72\text{e-}2 \rightarrow 2.23\text{e-}3$ A/cm) as L increases.
Oxide thickness t _{ox}	5.0e-7 → 2.0e-6 cm	V_TH strictly increases (0.78 → 1.34 V), I_ON strictly decreases ($1.01\text{e-}1 \rightarrow 1.63\text{e-}7$ A/cm).
Channel doping N_A	7.0e16 → 1.5e17 cm ⁻³	V_TH strictly increases (0.905 → 1.096 V).

I_OFF did not move meaningfully in any sweep — expected, since this idealized long-channel model has no DIBL/GIDL/punch-through mechanism. The doping sweep's upper bound (1.5e17, not the originally planned 1e18 cm⁻³) is a genuine numerical limit: higher doping sends the drift-diffusion Newton solver into non-convergent oscillation, since this model has no doping-dependent mobility degradation.

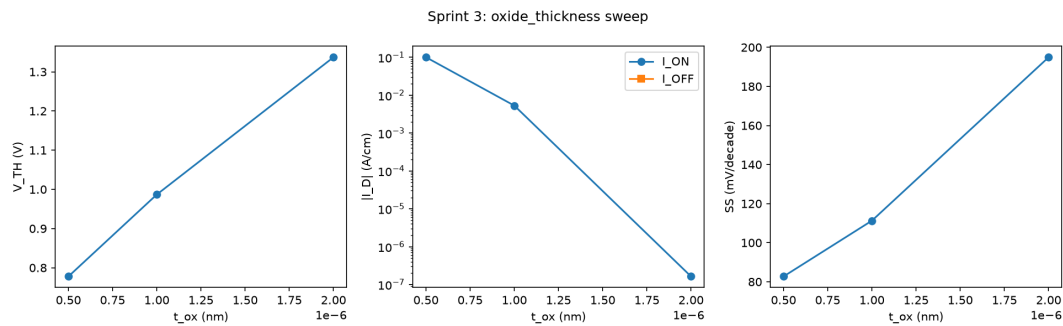


Figure 3. Oxide-thickness sweep: V_{TH} rises and I_{ON} falls monotonically as t_{ox} increases.

4.4 Low-power trade-off score

The 7 unique configurations were combined into one weighted score: $0.3 \times I_{ON} + 0.5 \times (1 - I_{OFF}) + 0.2 \times g_m$ (each term min-max normalized). Top by this score: thinnest oxide (t_{ox} = 5.0e-7 cm, score 0.846). Lowest: highest doping (N_A = 1.5e17, score 0.254).

Caveat, stated directly: I_{ON} spans roughly 6 decades across the 7 configurations while I_{OFF} spans under 1 decade — inside the solver's numerical noise floor (I_{OFF} even changes sign between configurations). So although I_{OFF} carries the largest weight (0.5), it is not resolving a real leakage difference; the ranking is effectively driven by I_{ON} and g_m. By the two metrics this dataset actually resolves, the best-supported "most conservative, lowest-drive" configuration is t_{ox} = 2.0e-6 cm (20 nm) — the global minimum in both I_{ON} and g_m — not the configuration the raw score ranks last.

4.5 Mesh sensitivity and reproducibility

Metric	Baseline mesh	2×-refined mesh	Rel. diff	Within 5%?
V _{TH} (V)	0.9867	0.9906	0.39%	PASS
g _{m,max} (A/cm/V)	0.06710	0.06672	0.56%	PASS
SS (mV/dec)	111.3	110.1	1.02%	PASS
I _{ON} (A/cm)	5.305e-3	5.001e-3	5.73%	FAIL (just over)
I _{OFF} (A/cm)	-6.08e-11	4.05e-11	166.6%	not meaningful (noise floor)

V_{TH}, g_m, and SS are mesh-converged well within tolerance. I_{ON} is not fully converged at the mesh density used throughout — every reported I_{ON} figure is accurate to roughly one part in twenty. A full clean-environment rebuild (.venv deleted and rebuilt from requirements.txt) reproduced the baseline device's results exactly, confirming the pinned dependencies give a deterministic solve.

5. Limitations, Stated Directly

None of these invalidate the qualitative findings above — the sweep trends are multi-decade effects, far larger than any of these bounds.

- **5.1 Scope:** 2D device only, drift-diffusion transport only (no quantum confinement, no ballistic/Monte Carlo transport), no fabrication-derived process variation, generic (not foundry-calibrated) parameters, SRH-only recombination, three parameters swept at three values each.
- **5.2 Device model:** ideal metal gate directly on the oxide — no gate work-function model, so flat-band voltage follows the structure's own built-in potential rather than a specific real gate material.
- **5.3 Numerical — I_ON precision:** baseline-mesh I_ON is accurate to about 5.7%, per the mesh-sensitivity check above.
- **5.4 Numerical — I_OFF noise floor:** off-state current sits near the solver's numerical floor across every sweep and the mesh check; comparisons of I_OFF between configurations are not physically meaningful below about one decade of spread. This directly weakens the low-power score's leakage term (§4.4).
- **5.5 Numerical — doping ceiling:** the channel-doping sweep is bounded at $1.5\text{e}17\text{ cm}^{-3}$ because higher doping causes non-convergent Newton-iteration oscillation in this model, absent a doping-dependent mobility model.

6. What Would Extend This Project

Explicitly out of scope for this project, but the natural next steps if the research question were extended:

- A short-channel leakage model (DIBL, GIDL, punch-through) to make I_OFF actually respond to the swept parameters — needed to answer the leakage side of the research question with confidence.
- A doping-dependent mobility model (e.g. Caughey-Thomas), to push the channel-doping sweep past its current $1.5\text{e}17\text{ cm}^{-3}$ numerical ceiling.
- A finer default mesh, to bring I_ON precision tighter than the current ~5.7% bound.
- A wider parameter sweep (more values per parameter, or a fourth parameter such as source/drain doping) once the above numerical limits are addressed.

7. Source Files, for Direct Reference

Full detail behind every claim in this document: docs/project_manual.md (scope/layout), docs/physics.md (equations/constants), docs/validation.md (every recorded result, by sprint), docs/optimization.md (trade-off method and full interpretation), docs/limitations.md (every scope, modeling, and numerical limitation found). A runnable, re-executable walkthrough of the same results is in notebooks/summary_walkthrough.ipynb.